St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank {Revised Curriculum (LOCF) - 2020-22 Batch }

Second Year –Third Semester

Course Title (Paper Title): Advanced Physical Chemistry.

QP Code: 85323

Unit	SI. No	Questions	Marks
1	1.	Write any four application of Enzyme kinetics.	02
1	2.	Draw Michaelis - Menton graph and mark $V_{max}/2$.	02
1	3.	Write Michaelis - Menton equation and explain it terms	02
1	4.	What is enzyme kinetics? Write Michaelis - Menton equation.	02
1	5.	What is Michaelis – Menton constant?	02
1	6.	Draw the graph to show the effect of substrate and hence mark V_{max} and $K_{m.}$	02
1	7.	Compare Michaelis – Menton and Line Weaver Burk plot.	02
1	8.	Draw Line Weaver Burk plot and hence write its significance	02
1	9.	Differentiate Simple enzyme and allosteric enzyme.	02
1	10.	A metal kept in vaccume does not undergo corrosion. Give reasons	02
1	11.	Draw a graph which show the effect of pH and temperature on the enzyme catalyzed reaction	02
1	12.	Draw a graph which show the effect of pH on the enzyme catalyzed reaction	02
1	13.	Draw a graph which show the effect of temperature on the enzyme catalyzed reaction	02
1	14.	Draw a graph which show the effect of inhibitor on the enzyme catalyzed reactio	02
1	15.	Write the application of glucose oxidase in the oxidation of glucose	02
1	16.	Write the reaction involved oxidation of glucose by glucose oxidase enzyme.	02
1	17.	Explain a kinetic applications of glucose oxidase	02
1	18.	What Hammett equation describe?	02
1	19.	What are the applications of Hammett equation?	02
1	20.	Why Hammett equation is called as linear free energy relationship?	02
1	21.	How do you draw a Hammett plot?	02
1	22.	What do you mean by linear free energy relationship?	02
1	23.	Why do ortho isomers and aliphatic compounds not exhibit the straight line relationship in Hammett equation?	02
1	24.	What is the importance of Taft equation?	02
1	25.	Write Taft equation and explain its terms.	02
1	26.	Write Hammett and Okemoto Brown equation and explain its terms.	02

1	**27.**	Write Taft and Okemoto Brown equation and explain its terms.	02
1	**28.**	Write the Swain-Scott equation and explain its terms	02
1	**29.**	Write the Edward equation and explain its terms	02
1	**30.**	Write the Winstein - Grunwald relationship and explain its terms	02
1	**31.**	What is Winstein - Grunwald relationship? Write its significance	02
1	**32.**	What are the significance Winstein - Grunwald relationship	02
1	**33.**	Write Edward equation and how it signify over Swain-Scott equation.	02
1	**34.**	What is meant by Exner criterion?	02
1	**35.**	What is meant by Isokinetic relationship?	02
1	**36.**	Write the significance of isokinetic temperature?	02
1	**37.**	What is meant by Isokinetic temperature?	02
1	**38.**	What is meant by kinetic isotope effect? Explain	02
1	**39.**	What happen when an element in a compound is replaced by an isotope of the element? Explain	02
1	**40.**	What happen to the rate of reaction when an ordinary hydrogen atom is replaced a deuterium (D) or tritium (T) atom? Explain	02
1	**41.**	What is meant by normal isotope effect? Explain	02
1	**42.**	What is meant by inverse isotope effect? Explain	02
1	**43.**	Differentiate between normal and inverse isotope effect.	02
1	**44.**	Write any two application of kinetic isotope effect.	02
1	**45.**	What is meant by Primary isotope effect?	02
1	**46.**	What is meant by secondary isotope effect?	02
1	**47.**	Differentiate between Primary and secondary isotope effect.	02
1	**48.**	What is meant by solvent isotope effect?	02
1	**49.**	Give any two difference between Primary and secondary isotope effect.	02
1	**50.**	Under most in vitro assay conditions, the enzyme is used in catalytic amounts (10 ^{12} to 10^{-8} M). Estimate the concentration of an enzyme in a living cell. Assume that (a) fresh tissue is 80% water and all of it is intracellular, (b) the total soluble protein in a cell represents 15% of the wet weight, (c) all the soluble proteins are enzymes, (d) the average molecular weight of a protein is 150,000 and (e) about 1000 different enzymes are present.	02
1	**51.**	Given the reaction $E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_p} E + P$ where $k_1 = 1 \times 10^7\ M^{-1}$ sec $k_{-1} = 1\times10^2\ sec^{-1}$ and $k_p = 3\times10^2\ sec^{-1}$, Calculate (a) K_s and (b) K_m	02
1	**52.**	(a) What fraction of V_{max} is observed at [S]=4 K_m, [S]=5 K_m, [S]=6 K_m, [S]=9 K_m and [S]=10 K_m? (b) Calculate the ratios of $[S]_{0.9}/[S]_{0.5}$.	02
1	**53.**	(a) What fraction of V_{max} is observed at [S]=4 K_m, [S]=5 K_m, [S]=6 K_m, [S]=9 K_m and [S]=10 K_m? (b) Calculate the ratios of $[S]_{0.75}/[S]_{0.5}$.	02
1	**54.**	Estimate k, the first-order rate constant, for an enzyme preparation with a V_{max} of 4.6 μ moles x x liter^{-1} x min^{-1} under the given experimental conditions. $K_m = 2\times 1$ ^{6} M.	02

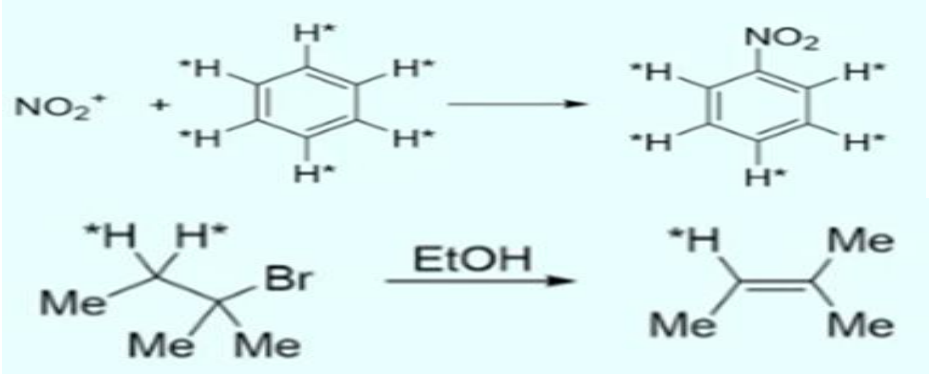
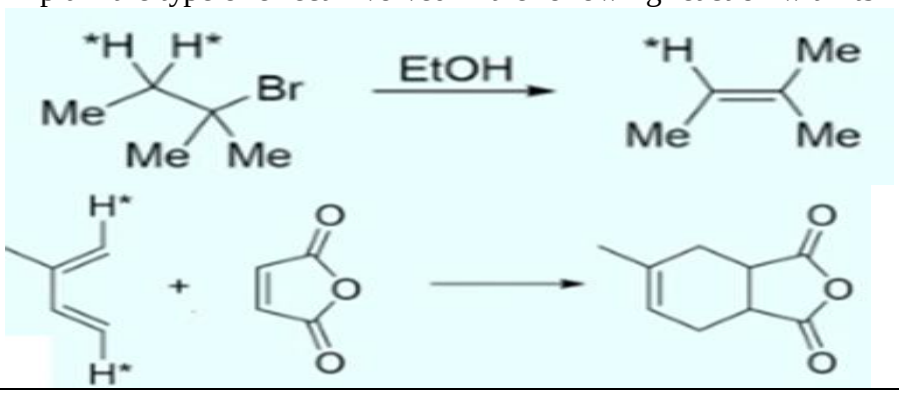
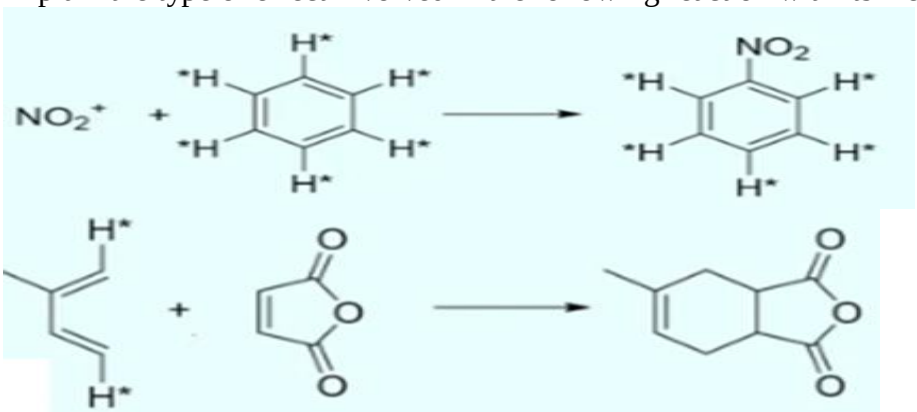
2	**55.**	Define half life and write a relation between half life and decay constant.	02
2	**56.**	What is Chain disintegration give example.	02
2	**57.**	Write an examples for alpha, beta, gamma decays.	02
2	**58.**	Write a decay scheme for an isotope $^{226}Ra_{88}$	02
2	**59.**	Define average life period	02
2	**60.**	Write decay kinetics equation and explain the terms involved in it.	02
2	**61.**	Define atomic mass unit.	02
2	**62.**	Define isotopes with example	02
2	**63.**	Define Mirror nuclei and isotones with examples	02
2	**64.**	What is Magic numbers.	02
2	**65.**	Define mass loss of a radioactive isotope.	02
2	**66.**	Define radioactive units Becquerel and curie	02
2	**67.**	Write the difference between transient and secular equilibrium,	02
2	**68.**	How to determine the half life for a mixture of independent half lives	02
2	**69.**	Define packing fraction.	02
2	**70.**	Define Binding energy and binding energy per nucleon.	02
2	**71.**	State Beer- Lamberts law and write its mathematical form	02
2	**72.**	Write the significance of term symbol.	02
2	**73.**	Define stability belt based on N/Z ratio.	02
2	**74.**	Define dead time and coincidence loss.	02
2	**75.**	Write the principle of G.M Counter.	02
2	**76.**	Define dead time of a GM Counter	02
2	**77.**	Write a note on decay kinetic equation and explain the terms involved in it.	02
2	**78.**	Define the terms dead time and coincidence loss	02
2	**79.**	Define half life and mean life of radioactive isotopes	02
2	**80.**	How does half life of a radio active nucleus is determined	02
2	**81.**	What do you mean by spallation reaction? Give one example.	02
2	**82.**	Give the significance of GM plateau	02
2	**83.**	Give the principle of Szilard-Chalmers process	02
2	**84.**	calculate the weight of 2milli curie of 14C ($t_{1/2}$ of ^{14}C= 5720yrs)	02
2	**85.**	Define radioactive decay	02
2	**86.**	A 0.1mg of sample of pure $^{231}Pu_{91}$ undergoes 1.04×10^{7} disintegration per minute.Calculate its half life	02
2	**87.**	Define half lifeand write a relation between half life and decay constant.	02
2	**88.**	Write the Bethe's notation for the general nuclear reaction.	02

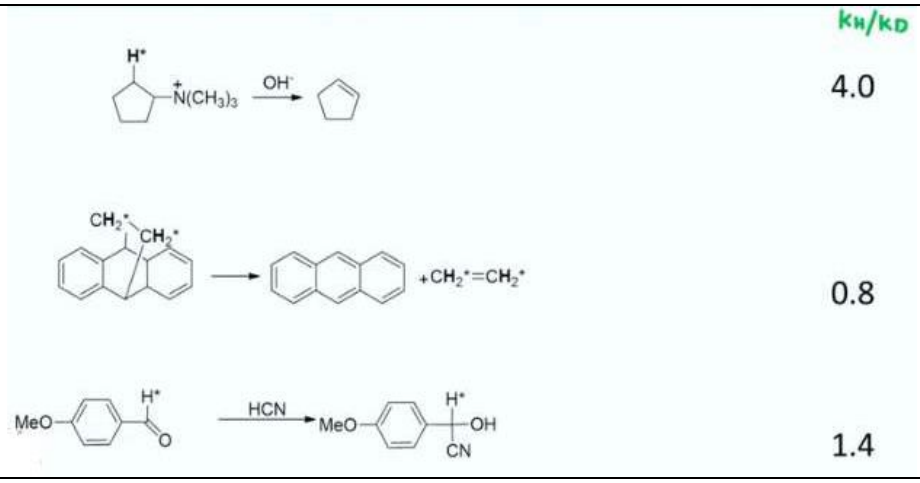
2	**89.**	S.T the binding energy and nuclear stability are related.	02
2	**90.**	What is radiation chemistry?	02
2	**91.**	How radiation interact with matters?	02
2	**92.**	Write one similarity and difference of X-rays and gamma rays.	02
2	**93.**	Compare the difference between X-rays and gamma-rays.	02
2	**94.**	Write the four principal mode of interaction of gamma radiation with matters	02
2	**95.**	What is Photoelectric effect? Write the equation for representing electron ejectin energy.	02
2	**96.**	Give the difference between θ and φ in Compton effect.	02
2	**97.**	Define (i) rad (ii) Gray	02
2	**98.**	Write the SI unit of dose in radiation energy	02
2	**99.**	What is RBE and Rem in radiation chemistry? Hence, Convert 1 sievert into rem.	02
2	**100.**	What is G value? Write the formula to calculate the G value and explain it's terms	02
2	**101.**	Define G value, How to denote G value for decomposition of $CaCO_3$	02
2	**102.**	Briefly explain the difference between radiation dosimetry and chemical dosimetry.	02
2	**103.**	Write the condition to satisfy by chemical dosimetry	02
2	**104.**	How to prepare Fricke dosimeter solution?	02
2	**105.**	Write the equation involved in Fricke dosimetry.	02
2	**106.**	What is Fricke dosimeter? Hence write the accepted G value of Fe^{+3}i.eG(Fe^{+3}).	02
2	**107.**	Write the value of pH when yield of H_2S is maximum and hence write the G (Cysteine) in radiolysis of cysteine amino acid.	02
2	**108.**	What happens to aqueous solution of cysteine when it irradiated for longer durati during radiolysis?	02
2	**109.**	Aromatic compounds are more stable than towards radiation than aliphatic compound Give reason with example.	02
2	**110.**	Write the reaction involved in radiolysis of biphenyl	02
2	**111.**	How radioisotope used in as tracers	02
2	**112.**	What is the principle for determination of solubility of sparingly soluble salt usin radioisotope tracer	02
2	**113.**	With equation of decay constant in carbon -14 dating and explain the term	02
2	**114.**	What are the two types of radioisotopes used in medical field?	02
2	**115.**	What is the use of Na-24 and I-131 radioisotope in detection purpose?	02
2	**116.**	What is specific activity, Also write principle of sparingly soluble salt determination using radioisotope.	02
2	**117.**	Mention any four fields where radioisotopes used as tracers	02
2	**118.**	Calculate the half life period of a radioisotope whose disintegration constant is 1.237×10^{-6} yrs	02
2	**119.**	What weight of ^{14}C will make one curie of it.($t_{1/2}$ = 5730yrs)	02
3	**120.**	Write the difference between homogeneous and heterogeneous photo catalysis.	02

3	**121.**	The transmittance of a solution of an organic compound in benzene at 301mμ at 25°C is 20% for a 2×10^{-3}M solution in a 1cm cell. Calculate the absorbance and molar extinction co-efficient.	02
3	**122.**	Absorption of each calorie gives 1.44×10^{-5}gm of hydrogen in the photo dissociation of HI vapor using radiation of wavelength 2070A°. Calculate the quantum yield.	02
3	**123.**	Calculate the energy produced/mole of photons having wavelength 2500A°.	02
3	**124.**	The extinction co-efficient of $Coen_2Br^{2+}$ is 40M/cm at 650mμ. Calculate the percent transmission for a 5cm cell filled with 0.01M	02
3	**125.**	Calculate the energy of one Einstein of a light of wavelength 3000A°	02
3	**126.**	Calculate the energy of Photon corresponding to wavelength 3600A°.	02
3	**127.**	State Beer- Lamberts law and write its mathematical form	02
3	**128.**	Write the significance of term symbol.	02
3	**129.**	Define quantum yield and mention its significance.	02
3	**130.**	Write the factors affecting quantum yield	02
3	**131.**	State Grothus-drapper law	02
3	**132.**	State Einstein law of photochemical equivalence.	02
3	**133.**	Define chain length and chain inhibition.	02
3	**134.**	Write the quantum yield and wavelength range for acetone and diethyl ketone actinometers.	02
3	**135.**	Define photosensitization. Give example	02
3	**136.**	Define the phenomena of phosphorescence.	02
3	**137.**	Define quenching and quencher	02
3	**138.**	Define the phenomena of fluorescence	02
3	**139.**	Write the applications of fluorescence	02
3	**140.**	Write an expression for photochemical rate law for the decomposition of acetaldehyde.	02
3	**141.**	Write the expression for photochemical rate law for the formation of hydrogen an chlorine.	02
3	**142.**	What is the effect of photo degradation on COD value.	02
3	**143.**	Write the principle involved in photo degradation.	02
3	**144.**	Define primary and secondary process in photochemical reaction.	02
3	**145.**	Write the different steps involved in photochemical reaction with example.	02
3	**146.**	Explain the principles of actinometers.	02
3	**147.**	Calculate the energy of one Einstein of a light of wavelength 3600°	02
3	**148.**	Define Photosensitization.	02
3	**149.**	What are term symbols	02
3	**150.**	Give the reasons for high quantum yield for the photochemical formation of HCl	02
3	**151.**	What is meant by COD	02

3	**152.**	Write the difference between acetone and diethyl ketone actinometers	02
3	**153.**	Write the term symbols for ^{19}F and ^{12}C.	02
3	**154.**	What is the effect of pressure on energetic of cell reaction	02
3	**155.**	What is the effect of temperature and concentration on energetics of cell reaction	02
3	**156.**	What are batteries and write its classification	02
3	**157.**	What are lithium batteries give example.	02
3	**158.**	How to determine the energy of electromagnetic radiation by using Planck's law.	02
3	**159.**	Define Einstein. Calculate the value of Einstein in cal/mole.	02
3	**160.**	State laws of light absorption.	02
3	**161.**	How to verify Beer's law	02
3	**162.**	State Laws of photochemistry	02
3	**163.**	State Stark-Einstein law	02
3	**164.**	Write the statement for principle of photochemical activation.	02
3	**165.**	Define the term photo catalysis.	02
3	**166.**	Write the mechanism for the production of OH radical by Fenton system.	02
3	**167.**	How heterogeneous photo catalysis differs from homogeneous photo catalysis.	02
3	**168.**	Define the term photo catalysts with examples.	02
3	**169.**	How chemical oxygen demand differ from biological oxygen demand.	02
3	**170.**	How photo degradation efficiency can be measured using COD value	02
3	**171.**	Define the term photo degradation.	02
3	**172.**	Define photochemical reactions with example.	02
3	**173.**	Write the mechanism involved in the photochemical combination of Hydrogen ar bromine	02
3	**174.**	Write the mechanism involved in the photochemical combination of Hydrogen ar chlorine	02
3	**175.**	Illustrate the steps involved in photochemical reactions with example.	02
3	**176.**	Write a note on chain inhibition with example.	02
3	**177.**	Discuss the two distinct primary and secondary process involved in photochemica reactions.	02
3	**178.**	What is study state approximation. How it is achieved.	02
3	**179.**	Write the steps involve in the photochemical decomposition of HI	02
3	**180.**	Define photosensitization and quenching with example	02
3	**181.**	How to represent the term symbol by using n,L,S and J and explain the terms involved in it.	02
3	**182.**	Calculate the term symbol for Nitrogen	02
3	**183.**	What is the term symbol for fluorine.	02
3	**184.**	Obtain a term symbol for Manganese.	02

3	**185.**	How acetaldehyde decomposes to give ethane.	02
1	**186.**	Write the assumption of Michaelis – Menton studies on hydrolysis of sucrose and hence draw Michaelis – Menton and Line Weaver Burk plot	05
1	**187.**	Describe a report on Line Weaver Burk reciprocal plot.	05
1	**188.**	What is the relative activity and the degree of inhibition caused by competitive inhibitor when [S] =K_m and [I]=K_i	05
1	**189.**	Explain the effect of pH on rate of corrosion of metal	05
1	**190.**	Explain the effect of temperature and pH on the rate of enzyme catalysed reaction	05
1	**191.**	Describe effect of temperature, Substrate and pH on the rate of enzyme catalyzed reaction.	05
1	**192.**	Write a note on effect of temperature and inhibitor on the rate of enzyme catalyze reaction.	05
1	**193.**	Discuss the effect of catalyst, pH and temperature on the rate of enzyme catalyzed reaction.	05
1	**194.**	Draw a graph which show the effect of different parameters on the enzyme catalyzed reaction.	05
1	**195.**	Write note kinetic and mechanistic application of glucose oxidase in the oxidation of glucose.	05
1	**196.**	Explain kinetic and mechanistic applications of glucose oxidase in the oxidation glucose.	05
1	**197.**	Describe the kinetic and mechanistic applications of glucose oxidase in the oxidation of glucose.	05
1	**198.**	Write a note on Hammett equation with an example.	05
1	**199.**	Explain the significance of Hammett equation.	05
1	**200.**	Write the Physical Meaning of σ and ρ and hence write the Uses of Hammett Plots.	05
1	**201.**	Explain how to draw Hammett plot using an example.	05
1	**202.**	What is Taft Equation? How to evaluate σ^* and Es?	05
1	**203.**	Write a note on Taft Equation and hence write σ^* and Es evaluation.	05
1	**204.**	Describe a report on Taft equation.	05
1	**205.**	Explain Okemoto Brown equation with an example.	05
1	**206.**	Write a note on a) Hammett equation b) Okemoto Brown equation	05
1	**207.**	Write a note on a) Hammett equation b) Taft equation	05
1	**208.**	Write a note on Swain-Scott equation and Winstein - Grunwald relationship	05
1	**209.**	Describe a report on with an example	05
1	**210.**	Explain the Winstein - Grunwald relationship with example	05
1	**211.**	Discuss and write brief note on the Swain-Scott equation and Edward equation.	05
1	**212.**	Explain Iso Isokinetic relationship and significance	05
1	**213.**	Write a brief note on Isokinetic relationship and significance of isokinetic temperature.	05
1	**214.**	Describe a brief report on a) Isokinetic relationship b) Exner criterion	05
1	**215.**	Explain the effect of reaction kinetics involving changes sp^2 to sp^3 and sp^3 to sp^2 and hence mention the type of effect involved.	05

1	**216.**	Write a note kinetic isotope effect.	05
1	**217.**	Describe a report on Primary and secondary isotope effect.	05
1	**218.**	Discuss the effect of reaction kinetics involving hybridization changes.	05
1	**219.**	Explain the type of effect involved in the following reaction with its mechanism. 	05
1	**220.**	Explain the type of effect involved in the following reaction with its mechanism. 	05
1	**221.**	Explain the type of effect involved in the following reaction with its mechanism. 	05
1	**222.**	Describe a report on solvent isotope effect.	05
1	**223.**	Write a note on a) Solvent isotope effect. b) Secondary isotope effect.	05
1	**224.**	What is secondary isotope effect? Match the K_H/K_D values of the following reaction and Give reasons.	05

			
1	**225.**	The following data were recorded for the enzyme-catalyzed reaction S → P. [S] (M) | v (nmoles x liter^{-1} x min) 6.25×10^{-6} | 15.0 7.50×10^{-5} | 56.25 1.00×10^{-4} | 60 1.00×10^{-3} | 74.9 1.00×10^{-2} | 75 (a) Estimate V_{max} and K_m. (b) What would v be at [S] =2.5×10^{-5} M and at [S]=5.0×10^{-5}M? (c) What would v be at 5.0×10^{-5}M, if the enzyme concentration were doubled? **(2+2+1)**	05
1	**226.**	The equilibrium constant for the reaction S → P is 5. Suppose we have a mixture of [S]=2×10^{-4} M and [P]=3×10^{-4} M. $K_{ms}=3\times10^{-5}$M, V_{maxf} = 2μ moles x liter^{-1} x min^{-1} V_{maxr} = 4 μ moles x moles x liter^{-1} x min^{-1}. (a) In which direction wi the reaction proceed on addition of an appropriate enzyme? (b) At what initial velocity will the reaction start towards equilibrium? **(1+4)**	05
1	**227.**	An enzyme was assayed at an initial substrate concentration of 10^{-5} M. The K_m fo the substrate is 2×10^{-3} M. At the end of 1 min, 2% of the substrate had been converted to product. (a) If the initial concentration of substrate were 10^{-6} M, wha percent of the substrate will be converted to product after 3 min? (b) What is the maximum attainable velocity (V_{max}) with the enzyme concentration used? **(1+4)**	05
1	**228.**	An enzyme was assayed at an initial substrate concentration of 10^{-5} M. The K_m fo the substrate is 2×10^{-3} M. At the end of 1 min, 2% of the substrate had been converted to product. (a) At about what substrate concentration will V_{max} be observed? (b) At this S concentration, what percent of the substrate will be converted to product in 3 min? **(1+4)**	05
1	**229.**	An enzyme was assayed at an initial substrate concentration of 2×10^{-5} M. In 6 min, half of the substrate had been used. The K_m for the substrate is 5×10^{-5}M. Calculate (a) k, (b) V_{max}, and (c) the concentration of product produced by 15 min. **(1+2+2)**	05
2	**230.**	Discuss the types of radioactive decays with examples.	05
2	**231.**	Write a note on decay schemes with typical examples	05
2	**232.**	Obtain an equation for decay kinetics for spontaneous radioactive decay.	05
2	**233.**	Obtain a parent daughter decay growth relationship when daughter nucleus is stable.	05
2	**234.**	Derive a general expression for the activity of a daughter nuclide with the decay constants λ_1 and λ_2.	05

2	**235.**	Differentiate between radioactive equilibrium and secular equilibrium.	05
2	**236.**	Differentiate between secular equilibrium and transient equilibrium.	05
2	**237.**	How do you determine the half lives for different radioactive nuclides.	05
2	**238.**	Explain the determination of half lives for a mixture of independent half-lives.	05
2	**239.**	Explain the determination of half-lives for a nuclides having moderately long and very short half-life.	05
2	**240.**	Explain the nuclear stability based on even-odd nature of the number of protons and neutrons.	05
2	**241.**	Explain the nuclear stability based on N/Z ratio.	05
2	**242.**	Calculate the binding energy/ nucleon of hydrogen atom in $^{1}H_{2}$ which has a mass of one proton 1.00782 a.m.u, mass of one neutron= 1.008665 a.m.u and mass of hydrogen atom = 2.014102 a.m.u	05
2	**243.**	Calculate the binding energy/ nucleon of Iron atom in $^{56}Fe_{26}$ which has a mass of one proton 1.00782 a.m.u, mass of one neutron= 1.008665 a.m.u and mass of hydrogen atom = 2.014102 a.m.u	05
2	**244.**	Define binding energy of nucleons and explain how it is related to the mass defec	05
2	**245.**	How do you separate radioisotopes by Szilard-Chalmersprocess.	05
2	**246.**	Describe the method of determining half life of a radioactive material.	05
2	**247.**	One gram of radioactive element decays by beta emission to 0.125g in 200hrs. How much more time will elapse until only 0.10g of it is left.	05
2	**248.**	Differentiate between elastic and inelastic scattering with examples.	05
2	**249.**	Write a note on Szilard-Chalmers process and Spallation reactions.	05
2	**250.**	Give the brief survey of alpha, beta, gamma decays.	05
2	**251.**	Write a note on nuclear stability based on N/Z ratio	05
2	**252.**	Write a note on Mass loss, Binding energy and binding energy per nucleon.	05
2	**253.**	Calculate the binding energy/nucleon of $^{2}He_{4}$	05
2	**254.**	Derive a general equation for the growth of a radioactive daughter. A-----B------C with decay constants λ_1 and λ_2	05
2	**255.**	Explain briefly about G.M Counters	05
2	**256.**	What is photonuclear and specific nuclear reactions	05
2	**257.**	Obtain an expression for parent-daughter decay growth relationship.	05
2	**258.**	Explain different types of nuclear reactions based on the nature of bombarding particles.	05
2	**259.**	Write a note on G.M.Plateau and dead time.	05
2	**260.**	Obtain an expression for the determination of dead time of a GM counter	05
2	**261.**	Obtain a relation between parent and daughter nuclei when half life of parent is greater than daughter	05
2	**262.**	What is binding energy? Calculate the binding energy/ nucleon in$^{235}U_{92}$atom whi has a mass of one proton 1.00720 a.m.u, mass of one neutron= 1.008665 a.m.u ar mass of hydrogen atom = 2,014102 a.m.u	05
2	**263.**	How is nuclear stability can be explained on the basis of binding energy and packing fraction.	05
2	**264.**	The half life of ^{22}Na is 2.6yr. 89% of it decays by β^+ emission and the rest by electron capture. Calculate Partial decay constant and half life for each mode.	05
2	**265.**	Explain the general characteristics of radioactive decay	05
2	**266.**	Differentiate between specific nuclear and photonuclear reactions.	05
2	**267.**	What is meant by dead time ? Explain how it is determined.	05
2	**268.**	What is meant by dead time of the GM counter? Explain the determination of dead time.	05

2	**269.**	Discuss the Oppenheimer-phillips process	05
2	**270.**	Discuss Photonuclear reactions.	05
2	**271.**	Write a note on mode of interaction of gamma radiation with matters.	05
2	**272.**	Explain Photoelectric effect after interaction between X-rays and matter	05
2	**273.**	Explain the dominant interaction of photons between 0.1 to 10 Mev.	05
2	**274.**	Describe the interaction of gamma radiation with matter using Compton effect.	05
2	**275.**	What is Compton wavelength? Write its equation, also, Explain the principle of pairs production in interacting of gamma radiation with matter.	05
2	**276.**	Write a note on(i) Pair production (ii) Compton scattering	05
2	**277.**	Describe the principle of gamma radiation interaction with matter in case of (i) Photoelectric effect (ii) Compton effect	05
2	**278.**	Explain briefly units of radiation energy.	05
2	**279.**	Write a note on (i) G value (ii) Units of radiation energy	05
2	**280.**	Write a note on Fricke dosimeter.	05
2	**281.**	Write a note on ceric-suphate dosimeter.	05
2	**282.**	Describe the preparation and procedure to reduce Ce^{+4} to Ce^{+3} ion using ceric sulphate dosimeter	05
2	**283.**	What is radiolysis? Also, explain what happens if radiolysis occurs in amino acid with an example	05
2	**284.**	Explain how thiol group of amino acid becomes self protective during radiolysis.	05
2	**285.**	Write the mechanism involved in radiolysis of cysteine	05
2	**286.**	Summarize a report on radiolysis of biphenyl.	05
2	**287.**	How radioisotopes used in elucidating the mechanism involved in Esterification reaction. reaction.	05
2	**288.**	How radioisotopes used in elucidating the mechanism involved in Friedal craft's reaction.	05
2	**289.**	Explain how radioisotopes used in elucidating the structure of PCl_5 and S_2O3^{2-}	05
2	**290.**	Illustrate elucidation of structural determination using radioisotope tracing technique	05
2	**291.**	Summarize a report on determination of reaction mechanism using radioisotope a tracer.	05
2	**292.**	Explain radioisotope tracer used in physicochemical problem of sparingly soluble substance.	05
2	**293.**	Explain neutron activation analysis	05
2	**294.**	Describe the role of C-14 isotope in the age determination.	05
2	**295.**	Write a note on use of radioisotopes in medical field.	05
2	**296.**	Summaries a report on hazards in radiochemical work.	05
2	**297.**	Explain the precautions taken in radiochemical work.	05
3	**298.**	Derive a photochemical rate law for the photochemical combination of hydrogen and chlorine	05
3	**299.**	Derive a photochemical rate law for the photochemical decomposition of acetaldehyde	05
3	**300.**	Write a note on potassium ferrioxalateactinometer.	05
3	**301.**	Explain about unaryl oxalate actinometer and the extent of decomposition of oxal acid titrated against $KMnO_4$	05
3	**302.**	Derive a term symbols for atomic carbon and fluorine.	05
3	**303.**	Discuss the mechanism involved in photo catalytic process	05
3	**304.**	Explain the applications of ZnO in photo degradation of dyes and pesticides.	05
3	**305.**	Write a note on photo catalysts ZnO and TiO_2.	05

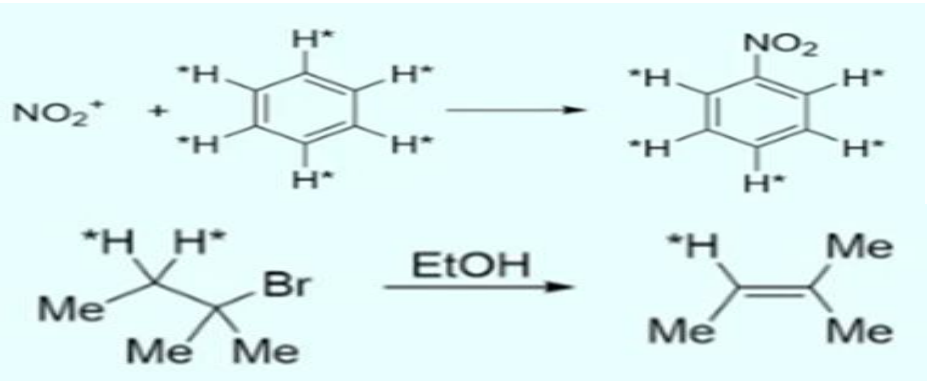
3	**306.**	What is an actinometer? Describe the functioning and application of Uranyl oxalate actinometers.	05
3	**307.**	Discuss the decomposition of acetaldehyde.	05
3	**308.**	Write a note on Effect of photo degradation on COD value and g value and factor affecting it.	05
3	**309.**	Write a note on dissociation of H_2 molecule by Photosensitization.	05
3	**310.**	Why ZnO/TiO_2 acts as photocatalyst in photo degradation of organic compounds	05
3	**311.**	Define fluorescence. Explain the characteristics and mechanism of quenching of fluorescence.	05
3	**312.**	Obtain the rate expression for the photochemical formation of HCl	05
3	**313.**	Discuss the photo degradation of a dye in the presence of TiO_2.	05
3	**314.**	Explain the resonance fluorescence and quenching of fluorescence	05
3	**315.**	What are primary and secondary batteries explain with examples.	05
3	**316.**	Write a note on lithium batteries	05
3	**317.**	What are storage batteries? Discuss the advantages and disadvantages of lithium storage batteries	05
3	**318.**	What are batteries? Discuss their classification and characteristic features.	05
3	**319.**	What is photo degradation? With illustrative example discuss the basic principle involved in the photo degradation of textile dyes using ZnO/ TiO_2.	05
3	**320.**	A radiation of $2530A^O$ incident on HI results in the decomposition of 1.85×10^{-2} o radiant energy. Calculate the quantum efficiency	05
3	**321.**	In the photochemical reaction B------C, 1.00×10^{-5} moles of C is formed as a resu of the absorption of 6.0×107 ergs at $3600A^o$. Calculate the quantum yield.	05
3	**322.**	A 0.003M solution of $Co(NH_3)_6^{3+}$ transmits 75% of incident light of 500mM if the path length is 1cm. Calculate the extinction co-efficient and the percent absorptio for a 0.01M solution.	05
3	**323.**	A 0.03M solution of $Co(C_2O_4)_3^{3-}$ has on O.D 2.0 at 660mM using a 1cm cell. Calculate the value of extinction Co-efficient and 1% absorption of a 0.015M solution in the same cell.	05
3	**324.**	When gaseous HI is illuminated with radiation of $2530A^O$ it is observed that 1.85 10^{-2} mole decomposes/1000cal of radiant energy absorbed. Calculate the quantum yield.	05
3	**325.**	Explain the mechanism of photosensitization of dissociation of hydrogen by mercury.	05
3	**326.**	Explain the mechanism involved in the determination of intensity of absorbed lig by the decomposition of oxalic acid by titration with $KMnO_4$.	05
3	**327.**	State and verify laws of light absorption.	05
3	**328.**	How to determine the quantum yield by determining the number of moles of substance reacting	05
3	**329.**	How to determine the quantum yield by determining the number of Einstein of light absorbed.	05
3	**330.**	Discuss the role of ZnO and TiO_2 in photo catalytic process.	05
3	**331.**	Discuss the mechanism involved in photo catalytic process with diagrammatic representation.	05
3	**332.**	Discuss the application of ZnO/TiO_2 in photo degradation of dyes.	05
3	**333.**	Write the applications of photo degradation of pesticides.	05
3	**334.**	State and obtain photochemical rate law by using study state approximation concept.	05
3	**335.**	Obtain an equation for rate law for the photochemical decomposition of hydroger iodide.	05

3	**336.**	Discuss the mechanism involved in the process of photosensitization.	05
3	**337.**	Discuss the rules involved in deriving term symbol for the ground states of atoms	05
3	**338.**	Explain briefly the applications of fluorescence in various fields.	05
3	**339.**	Discuss Rice-Herzfeld mechanism for the decomposition of acetaldehyde.	05
******			

*****

Question Paper Pattern- Model Question Paper Q.P Code: 85323

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE			
III SEMESTER –COURSE M.SC FINAL EXAMINATION			
SUBJECT : CHEMISTRY			
TITLE: ADVANCED PHYSICAL CHEMISTRY(HC)			
TIME: 3 HRS			**MAX MARKS: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	Draw the graph to show the effect of substrate and hence mark V_{max} and K_m	2
	b	Define Mirror nuclei and isotones with examples	2
	c	Calculate the weight of 2milli curie of ^{14}C ($t_{1/2}$ of ^{14}C= 5720yrs)	2
	d	What Hammett equation describe?	2
	e	How radiation interact with matters?	2
	f	Write decay kinetics equation and explain the terms involved in it.	2
	g	Write the application of glucose oxidase in the oxidation of glucose	2
	h	Give the significance of GM plateau	2
	i	What is RBE and Rem in radiation chemistry? Hence, Convert 1 sievert into rem.	2
	j	Write the Swain-Scott equation and explain its terms	2
	k	Write the value of pH when yield of H_2S is maximum and hence write the G	2
	l	Write the reaction involved in radiolysis of biphenyl	2
		Answer any Five of the following:	**5 x 10= 50**
2	a	Obtain a parent daughter decay growth relationship when daughter nucleus is stable.	5
	b	Calculate the binding energy/ nucleon of hydrogen atom in $^1H_2$ which has a mass of one proton 1.00782 a.m.u, mass of one neutron= 1.008665 a.m.u and mass of hydrogen atom = 2.014102 a.m.u	5
3	a	Explain Photoelectric effect after interaction between X-rays and matter	5
	b	What is radiolysis? Also, explain what happens if radiolysis occurs in amino acid with an example	5

4	a	Describe the role of C-14 isotope in the age determination.	5
	b	Derive a photochemical rate law for the photochemical combination of hydrogen and chlorine	5
5	a	Define fluorescence. Explain the characteristics and mechanism of quenching of fluorescence.	5
	b	An enzyme was assayed at an initial substrate concentration of 2×10^{-5} M. In 6 min, half of the substrate had been used. The K_m for the substrate is 5×10^{-5}M. Calculate (a) k, (b) V_{max}, and (c) the concentration of product produced by 15 min	5
6	a	Write a note on a) Solvent isotope effect. b) Secondary isotope effect.	5
	b	Write the Physical Meaning of σ and ρ and hence write the Uses of Hammett Plots.	5
7	a	Explain the type of effect involved in the following reaction with its mechanism. 	5
	b	Derive a general expression for the activity of a daughter nuclide with the decay constants λ_1 and λ_2.	5
8	a	One gram of radioactive element decays by beta emission to 0.125g in 200hrs. How much more time will elapse until only 0.10g of it is left.	5
	b	What is Taft Equation? How to evaluate σ* and Es?	5